# MSG No. 215.70 $P\bar{4}3m$ [ Type I, cubic ]

Table 1: Wyckoff site: `1a`, site symmetry: `-43m`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | `[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]` |

Table 2: Wyckoff site: `1b`, site symmetry: `-43m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]` |

Table 3: Wyckoff site: `3c`, site symmetry: `-42.m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,3,4,13,14,21,22]` |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[5,8,10,12,17,18,19,20]` |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[6,7,9,11,15,16,23,24]` |

Table 4: Wyckoff site: `3d`, site symmetry: `-42.m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | `[1,2,3,4,13,14,21,22]` |
| 2 | $\left[0, \frac{1}{2}, 0\right]$ | `[5,8,10,12,17,18,19,20]` |
| 3 | $\left[0, 0, \frac{1}{2}\right]$ | `[6,7,9,11,15,16,23,24]` |

Table 5: Wyckoff site: `4e`, site symmetry: `.3m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, x]$ | `[1,5,6,20,22,24]` |
| 2 | $[x, -x, -x]$ | `[2,10,11,16,17,21]` |
| 3 | $[-x, x, -x]$ | `[3,7,12,13,18,23]` |
| 4 | $[-x, -x, x]$ | `[4,8,9,14,15,19]` |

Table 6: Wyckoff site: `6f`, site symmetry: `2.mm`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | `[1,2,21,22]` |
| 2 | $[-x, 0, 0]$ | `[3,4,13,14]` |
| 3 | $[0, x, 0]$ | `[5,12,18,20]` |
| 4 | $[0, 0, x]$ | `[6,9,15,24]` |
| 5 | $[0, 0, -x]$ | `[7,11,16,23]` |
| 6 | $[0, -x, 0]$ | `[8,10,17,19]` |

Table 7: Wyckoff site: `6g`, site symmetry: `2.mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[1,2,21,22]` |
| 2 | $\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[3,4,13,14]` |
| 3 | $\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$ | `[5,12,18,20]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$ | `[6,9,15,24]` |
| 5 | $\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$ | `[7,11,16,23]` |
| 6 | $\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$ | `[8,10,17,19]` |

Table 8: Wyckoff site: `12h`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ 0\right]$ | `[1,2]` |
| 2 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | `[3,4]` |
| 3 | $\left[0,\ x,\ \frac{1}{2}\right]$ | `[5,12]` |
| 4 | $\left[\frac{1}{2},\ 0,\ x\right]$ | `[6,9]` |
| 5 | $\left[\frac{1}{2},\ 0,\ -x\right]$ | `[7,11]` |
| 6 | $\left[0,\ -x,\ \frac{1}{2}\right]$ | `[8,10]` |
| 7 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | `[13,14]` |
| 8 | $\left[0,\ \frac{1}{2},\ x\right]$ | `[15,24]` |
| 9 | $\left[0,\ \frac{1}{2},\ -x\right]$ | `[16,23]` |
| 10 | $\left[\frac{1}{2},\ -x,\ 0\right]$ | `[17,19]` |
| 11 | $\left[\frac{1}{2},\ x,\ 0\right]$ | `[18,20]` |
| 12 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | `[21,22]` |

Table 9: Wyckoff site: `12i`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ z]$ | `[1,20]` |
| 2 | $[x,\ -x,\ -z]$ | `[2,17]` |
| 3 | $[-x,\ x,\ -z]$ | `[3,18]` |
| 4 | $[-x,\ -x,\ z]$ | `[4,19]` |
| 5 | $[z,\ x,\ x]$ | `[5,24]` |
| 6 | $[x,\ z,\ x]$ | `[6,22]` |
| 7 | $[-x,\ z,\ -x]$ | `[7,13]` |
| 8 | $[-z,\ -x,\ x]$ | `[8,15]` |
| 9 | $[-x,\ -z,\ x]$ | `[9,14]` |
| 10 | $[z,\ -x,\ -x]$ | `[10,16]` |
| 11 | $[x,\ -z,\ -x]$ | `[11,21]` |
| 12 | $[-z,\ x,\ -x]$ | `[12,23]` |

Table 10: Wyckoff site: `24j`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 2 | $[x, -y, -z]$ | [2] |
| 3 | $[-x, y, -z]$ | [3] |
| 4 | $[-x, -y, z]$ | [4] |
| 5 | $[z, x, y]$ | [5] |
| 6 | $[y, z, x]$ | [6] |
| 7 | $[-y, z, -x]$ | [7] |
| 8 | $[-z, -x, y]$ | [8] |
| 9 | $[-y, -z, x]$ | [9] |
| 10 | $[z, -x, -y]$ | [10] |
| 11 | $[y, -z, -x]$ | [11] |
| 12 | $[-z, x, -y]$ | [12] |
| 13 | $[-x, z, -y]$ | [13] |
| 14 | $[-x, -z, y]$ | [14] |
| 15 | $[-z, -y, x]$ | [15] |
| 16 | $[z, -y, -x]$ | [16] |
| 17 | $[y, -x, -z]$ | [17] |
| 18 | $[-y, x, -z]$ | [18] |
| 19 | $[-y, -x, z]$ | [19] |
| 20 | $[y, x, z]$ | [20] |
| 21 | $[x, -z, -y]$ | [21] |
| 22 | $[x, z, y]$ | [22] |
| 23 | $[-z, y, -x]$ | [23] |
| 24 | $[z, y, x]$ | [24] |